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Drill Guide System

Abstract

An edge guide attachment for a drill guide having a base having a bottom defining a bottom plane includes a first mounting arm, a fence, and a ruler. The first mounting arm is configured to mount to the base of the drill guide. The fence has a workpiece abutment surface extending below the bottom plane when mounted to the drill guide such that the workpiece abutment surface can engage an edge of the workpiece when the base rests on a surface of the workpiece. The fence is adjustably positionable along the first mounting arm. The ruler is fixably attachable to the base and has measuring demarcations thereon for indicating a spacing of the abutment surface from a predetermined location of the drill guide. The fence is movable relative to the ruler when the ruler is attached to the base.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS [0001] This Patent application claims the benefit of U.S. Provisional Patent Application No. 63/559,585, filed Feb. 29, 2024, the entire teachings and disclosure of which are incorporated herein by reference thereto.

FIELD OF THE INVENTION

[0002] This invention generally relates to a drill guide and an edge guide attachment for use with a drill guide.

BACKGROUND OF THE INVENTION

[0003] Drill guides are used for precisely and repeatedly drilling holes within one or more workpieces. The drill guide acts to guide the drill bit as the user plunges the drill bit into the workpiece. The drill guide helps remove error that may occur when the drill bit is simply being guided by the user, e.g. if the user is providing all of the guiding of the drill and drill bit by hand.

[0004] Further, a drill guide can be used to prevent over drilling or drilling too far into a workpiece by providing travel limits, e.g. depth stops.

[0005] A drill guide may often be used to drill one or more holes in a workpiece at a desired location in a workpiece. For example, it may be desirable to form multiple holes at a same offset from an edge of a workpiece.

[0006] The present invention provides improvements over the current state of the art in drill guide systems.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The accompanying drawings incorporated in and forming a part of the specification illustrate several aspects of the present invention and, together with the description, serve to explain the principles of the invention. In the drawings:

[0008] FIG. 1 is a perspective illustration of a drill guide system;

[0009] FIG. 2 is a partial exploded illustration of the drill guide system having the edge guide attachment removed and in exploded form;

[0010] FIG. 3 is an enlarged exploded illustration of the edge guide attachment;

[0011] FIG. 4 is a bottom perspective illustration of the drill guide system of FIG. 1 with the edge guide attachment detached from the base of the drill guide; and

[0012] FIG. 5 is a cross-sectional illustration of the drill guide system of FIG. 1.

[0013] While the invention will be described in connection with certain preferred embodiments, there is no intent to limit it to those embodiments. On the contrary, the intent is to cover all alternatives, modifications and equivalents as included within the spirit and scope of the invention as defined by the appended claims.

DETAILED DESCRIPTION OF THE INVENTION

[0014] FIG. 1 illustrates a drill guide system **100**. The drill guide system **100** includes a drill guide **102** and an edge guide attachment **104** attached to the drill guide **102**. The drill guide system **100** is configured for performing accurate drilling of a work piece. In particular, the drill guide system **100** is configured for repeatable and accurate drilling depth and position.

[0015] The drill guide **102** is configured to allow a user to perform controlled and precise drilling of the workpiece. In particular, the drill guide **102** is used to control the drilling motion about a

drilling axis **108**.

[0016] The drill guide **102** has a base **110** that has a bottom that generally defines a bottom plane. In use, the base **110** may rest on the workpiece to be drilled or on a workbench or other object.

[0017] A pair of guides **114**, **116** are operably attached to the base **110**. A slide member **120** is carried by the guides **114**, **116** for sliding motion along the guides **114**, **116** towards or away from the base **110** as illustrated by arrow **122**.

[0018] In this example, the slide member **120** has a drill chuck **124** mounted thereto that can operably hold a drill bit or other boring bit. The drill chuck **124** is configured to rotate about drilling axis **108**. The drill chuck **124** has an input end **126** that can be attached to a power tool such as a drill to provide an input to drive the drill chuck **124**. For example, the input end **126** could be engaged by a chuck of an external drill for transferring torque from the external drill to the drill chuck **124**.

[0019] The slide member **120** extends laterally between the guides **114**, **116**.

[0020] In this example, the guides **114**, **116** straddle the drill axis **108** and the drill chuck **124**.

[0021] In FIG. **1**, the guides **114**, **116** and slide member **120** are oriented relative to the base **110** and its base plane such that the drill axis **108** is orthogonal to the base plane and thus would be orthogonal to a planar workpiece.

[0022] However, the drill guide **102** is configured such that the guides **114**, **116** may be pivoted about axis **130** relative to base **110** so as to alter the drilling angle relative to a workpiece (i.e. relative to the base plane), see e.g. arrow **132**. In some embodiments, this angle may be 60 degrees from orthogonal to the base plane about axis **130** for a total variation of approximately 120 degrees. However, other amounts and no amounts of angular adjustment are contemplated.

[0023] During drilling operations, the user will typically push the slide member **120** towards base **110** using a plunging motion. The guides **114**, **116** will accurately control the plunging motion to provide accurately located and formed holes in the workpiece.

[0024] A depth stop arrangement **140** is provided so that the depth of the plunging motion can be limited to prevent over penetration of the drilling motion into the workpiece and/or to provide for repeated depth drilling.

[0025] In this example, the depth stop arrangement **140** is not provided by the guides **114**, **116**. In some prior implementations, a depth stop was mounted to one of the guides **114**, **116** that would limit the sliding motion of the slide member **120** along the guide **114**, **116**. The user would simply slide the depth stop along the corresponding guide **114**, **116** to limit motion. Unfortunately, this resulted in compression of the corresponding spring **142** carried by the shaft **144** of the guide **114**, **116**. However, by doing this, this created a non-uniform loading on the slide member **120** by the springs **142** of the pair of guides **114**, **116** which resulted in generating a torque on the slide member **120** such that the drill chuck **124** would tilt relative to the shafts **144** reducing the accuracy of the drilling operation.

[0026] In this example, the depth stop arrangement **140** includes a separate stop rod **150** and depth stop collar **152** that is movable axially along the stop rod **150**. Here, the depth stop collar **152** does not interfere or interact with springs **142** and eliminates a problem with prior art depth stop arrangements.

[0027] In use, an abutment surface **156** of the slide member **120** will contact abutment surface **158** of the depth stop collar **152** when the user has plunged the slide member **120** the desired distance along axis **108** (e.g. formed the desired depth drilling action).

[0028] In this example, the stop rod **150** is mounted in fixed, parallel relation relative to shafts **144** of the guides **114**, **116**. However, the slide member **120** may move relative to stop rod **150**. In this example, end **160** of stop rod **150** is threaded into shaft base **162** that pivotally attaches the guides **116** to base **110**.

[0029] The opposite end **162** of stop rod **150** is slidably received in slide aperture **164** formed in slide member **120**. The slide aperture **164** allows the slide member **120** to slide relative to stop rod

150.

[0030] Adjustment of the position of the stop collar **152** along stop rod **150** adjusts and limits the amount of sliding motion of slide member **120** along the guides **114**, **116**.

[0031] Again, by having the stop arrangement **140** independent of the springs **142** and guides **114**, **116**, the slide member **120** is not exposed to uneven forces and thus torques as it is transitioned towards base **110**.

[0032] In addition to accurate plunging, the drill guide system **100** provides for accurate location relative to an edge of a workpiece. In particular, an edge guide attachment **104** is provided that can locate the drill guide **102** relative to the edge of a workpiece.

[0033] The edge guide attachment **104** generally includes at least one and typically two mounting arms **161**, **163**, a fence **166** and a ruler **168**.

[0034] In this example, the edge guide attachment **104** is removable attachable to the drill guide **102** and particularly to the base **110**.

[0035] Here, the mounting arms **161**, **163** thread into base **110**.

[0036] Flats on the sides of the mounting arms **161**, **163** allow for engagement with a wrench or other tool for assisting in threadedly attaching the mounting arms **161**, **163** to the base **110**.

[0037] The mounting arms **161**, **163** are generally parallel to offset axis **170**.

[0038] The fence **166** is adjustably positionable along the mounting arms **161**, **163**. Locking members in the form of locking screws **171** can fix the position of the fence **166** relative to mounting arms **161**, **163**. Other locking members such as pins or catches could be provided.

[0039] The fence **166** includes a workpiece abutment surface **172** that extends below the bottom plane (e.g. bottom **112**) of the base **110**. As such, when the base **110** rests on a workpiece, the workpiece abutment surface **172** will be able to engage an edge or side of the workpiece.

[0040] Ruler **168** allows the user to precisely locate fence **166** and particularly workpiece abutment surface **172** relative to the drill guide **102** and particularly relative to a predetermined location of the drill guide **102**. In this example, the predetermined location is the drill axis **108** and particularly the drill axis **108** when it is oriented orthogonal to the base plane. The distance D between the drill axis **108** and the workpiece abutment surface **172** is illustrated in FIG. 5.

[0041] The ruler **168** includes measuring demarcations **180**, **182** for identifying the offset of the workpiece abutment surface **172** from the drill axis **108**. Thus, if the user aligns the workpiece abutment surface **172** with the desired offset for drilling relative to the edge of the workpiece as represented by the measuring demarcations **180**, **182** a hole drilled using the drill guide will be offset from the edge of the workpiece that amount.

[0042] In some embodiments, the demarcations may provide both standard and metric units. In some embodiments only one measurement unit, e.g. only standard or only metric units may be provided. The demarcations may be machined into the surface of the ruler or printed on the ruler.

[0043] In this example, the ruler **168** is positioned laterally between the mounting arms **161**, **163**.

[0044] The fence **166** is an L-shaped member having a foot **184** and an upstanding wall **186**. The upstanding wall **186** provides workpiece abutment surface **172**. In other examples, the foot **184** may be eliminated.

[0045] The upstanding wall **186** includes a ruler receiving aperture **190** through which a free end of the ruler **168** extends when assembled. In this example, the ruler receiving aperture **190** extends through the workpiece abutment surface **172**.

[0046] Similarly, mounting arm apertures **194**, **195** extend through the upstanding wall **186** and slidably receive the mounting arms **161**, **163**. The locking screws **171** extend through a top end of the upstanding wall **186** and selectively engage and release the mounting arms **161**, **163** for allowing for adjustment of the position of the fence **166**.

[0047] The base **110** includes a mounting slot **200** that receives an end of the ruler **168**. Screws **202** releasably attach the ruler **168** to the base **110**. In this example, slot **200** extends through a vertical side **210** of base **110**. The use of the slot **200** helps maintain the proper orientation for ruler **168** to

base **110**.

[0048] FIGS. **1** and **5** illustrate the fence **166** mounted to the mounting arms **161**, **163** in a first orientation with foot **184** extending away from the upstanding wall **186** in a direction that extends away from the base **110**. This orientation can be used when the hole to be drilled into a workpiece is sufficiently far from an edge of the workpiece that the base **110** would not extend outward beyond the adjacent edge of the workpiece. In other words, the spacing of the hole to be drilled from the edge is greater than the spacing from axis **108** to the side of the base through which slot **200** extends.

[0049] FIGS. **2** and **4** illustrate the fence **166** in a second orientation (FIG. **4** actually mounted). In this second orientation, the foot **184** extends from the upstanding wall **186** towards the base **110**. This orientation can be used when the hole to be drilled into the workpiece is too close to the edge of the workpiece that the base **110** must extend outward beyond the edge of the workpiece. Here, foot **184** may extend under the base **110** so as to abut the edge of the workpiece that the base extends outward beyond. In other words, the spacing of the hole to be drilled from the edge of the workpiece is less than the spacing from axis **108** to the side of the base **110**.

[0050] The free end **212** of foot **184** provides a second abutment surface that can abut the edge of the workpiece to properly orient the drill guide **102** relative to the edge of the workpiece.

[0051] When oriented in this second orientation, either 1) the ruler **168** has a set of demarcations that correspond to this configuration or the user must subtract the width **W1** from the demarcations **180**, **182** so as to know the proper spacing of the axis **108** from the abutted edge of the workpiece.

[0052] Thus, in one embodiment, the first set of demarcations **180** may correspond to the offset when using the first orientation as illustrated in FIG. **1** and the second set of demarcations **182** may be used in the second orientation illustrated in FIG. **4**.

[0053] All references, including publications, patent applications, and patents cited herein are hereby incorporated by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

[0054] The use of the terms “a” and “an” and “the” and similar referents in the context of describing the invention (especially in the context of the following claims) is to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The terms “comprising,” “having,” “including,” and “containing” are to be construed as open-ended terms (i.e., meaning “including, but not limited to,”) unless otherwise noted. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the invention.

[0055] Preferred embodiments of this invention are described herein, including the best mode known to the inventors for carrying out the invention. Variations of those preferred embodiments may become apparent to those of ordinary skill in the art upon reading the foregoing description. The inventors expect skilled artisans to employ such variations as appropriate, and the inventors intend for the invention to be practiced otherwise than as specifically described herein.

Accordingly, this invention includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the invention unless otherwise indicated herein or otherwise clearly contradicted by context.

Claims

1. An edge guide attachment for a drill guide, the drill guide having a base having a bottom defining a bottom plane, the edge guide attachment comprising: a first mounting arm configured to mount to the base; a fence having a workpiece abutment surface extending below the bottom plane when mounted to the drill guide such that the workpiece abutment surface can engage an edge of the workpiece when the base rests on surface of the workpiece, the fence adjustably positionable along the first mounting arm; a ruler fixably attachable to the base having measuring demarcations thereon for indicating a spacing of the abutment surface from a predetermined location of the drill guide, the fence being movable relative to the ruler when the ruler is attached to the base.
 2. The edge guide attachment of claim 1, further comprising a second mounting arm configured to mount to the base in parallel relation to the first mounting arm, the fence being movable relative to the second mounting arm.
 3. The edge guide attachment of claim 2, wherein the ruler is located between the first and second mounting arms when assembled.
 4. The edge guide attachment of claim 1, wherein the workpiece abutment surface has an aperture therethrough, the ruler extending through the aperture when assembled.
 5. The edge guide attachment of claim 2, wherein the first and second mounting arms have threaded ends configured to threadedly mate with the base to secure the first and second mounting arms to the base.
 6. The edge guide attachment of claim 2, wherein the first and second mounting arms extend through corresponding first and second apertures through the workpiece abutment surface when assembled.
 7. The edge guide attachment of claim 6, further comprising first and second locking members configured to releasably lock the fence to the first and second mounting arms to selectively fix the position of the fence along the first and second mounting arms.
 8. A drill guide system comprising: a drill guide including: a base having a bottom defining a bottom plane; first and second guides operably attached to the base; a slide member mounted to the guides for sliding motion along the first and second guides towards and away from the base; an edge guide attachment of claim 1.
 9. The system of claim 8, wherein the base includes a mounting slot receiving an end of the ruler therein, when assembled.
 10. The system of claim 8, wherein when the edge guide attachment is assembled with the base, the ruler remains fixed relative to the base as the fence is moved along the first mounting arm relative to the base.
 11. The system of claim 8, wherein the chuck assembly defines a drilling axis and the spacing is the spacing between the drilling axis when it is oriented orthogonal to the bottom plane and the workpiece abutment surface.
 12. The system of claim 8, wherein the slide member has a drill chuck mounted thereto for rotation relative to the slide member about a chuck axis, the drill chuck having an input end that is engageable by the chuck of an external drill for driving the drill chuck for rotation about the chuck axis with the external drill.
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